

Watson Hartsoe

Research engineer and creative technologist · PhD candidate, Digital Media, Georgia Tech

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SUMMARY

I build working tools for generative media, filmmaking and worldbuilding. Machines now produce more images, video, code and possible worlds than people can inspect, organize or shape; my work turns that abundance into usable instruments: browser-based filmmaking systems, networked 3D environments, and compilers that translate formal languages into interactive experiences. I work by building the system, testing it with real users, watching where it fails, and revising it.

EDUCATION

PhD candidate, Digital Media Georgia Institute of Technology

School of Literature, Media, and Communication. Research on how people work with generative systems, through a method called thick prompting.

MS, Human-Computer Interaction Georgia Institute of Technology

Fulbright research, ethnographic fieldwork Honduras

Arts-based and participatory research with communities in Central America.

SELECTED WORK

CINEOSIS Tool

A browser-based instrument for assembling abundant generated footage: assembly as sampling, not cutting.

Where You Go When You Leave Feature film, in production

With poet and filmmaker Mark Anthony Thomas: carrying a poet's language into a generative medium.

The Machinery of Meaning Film engine

A procedural film engine you can read while it runs; every diagram, transition and claim comes from one inspectable timeline.

Can a Model Build with LEGO? Interactive essay

Translates a language model's claims into legal LDraw parts, exact placements and real connections that can be checked.

Centaur Box Talk

Exposes the hidden run record beneath a persuasive human–AI conversation.

Gumball Emotion Machine Physical computing

Reads a face, assigns an emotion, returns a gumball: emotion classification made physical.

The Grove 340 repositories

An index of every project: games, tools, 3D worlds and film systems, each assessed and previewed.

PUBLICATIONS AND TALKS

- Auditing Emotion AI. NordiCHI 2024. Documents where emotion-recognition systems mislabel affect.
- Coaxing the Ripples. Electronic Literature Organization (ELO) 2026. Can a generative system remember what already happened in its world?
- Operative Ekphrasis: From Imagetext to Worldtext. With Jay David Bolter.
- After the Scene. The LEGOS framework for what a model writes after a scene feels finished.
- Growing Entanglements. Captain Cook and seven agent-based models of a historical encounter.

RECOGNITION

- ITU AI for Good platform
- Georgia Tech AI Competition
- CREATE-X, Georgia Tech
- Neu Wave AI Film Festival
- Fulbright

SKILLS

Interfaces and graphics	JavaScript, HTML and CSS; three.js and WebGL; GLSL shaders; Canvas and SVG; Web Audio; responsive, touch-first and accessible interfaces (ARIA).
AI systems	Large language model interfaces and APIs; prompt and context engineering; computer vision and speech in the browser; evaluation of model behaviour.
Networked media	WebRTC and WebSockets; browser-based film and media pipelines; static publishing on GitHub Pages.
Research methods	Prototyping and iterative design; user testing; ethnographic fieldwork; participatory design with museums and youth programs; thick prompting.

EARLIER WORK

Participatory worldmaking with museums and youth programs: museum collections as social VR spaces (Hubs Cloud), environmental futures as web VR worlds (Future Oceans), civic futures prototyping (Future Scouts, Leimert Phone Company), and youth photography as community narrative in Honduras (Ojo Joven).